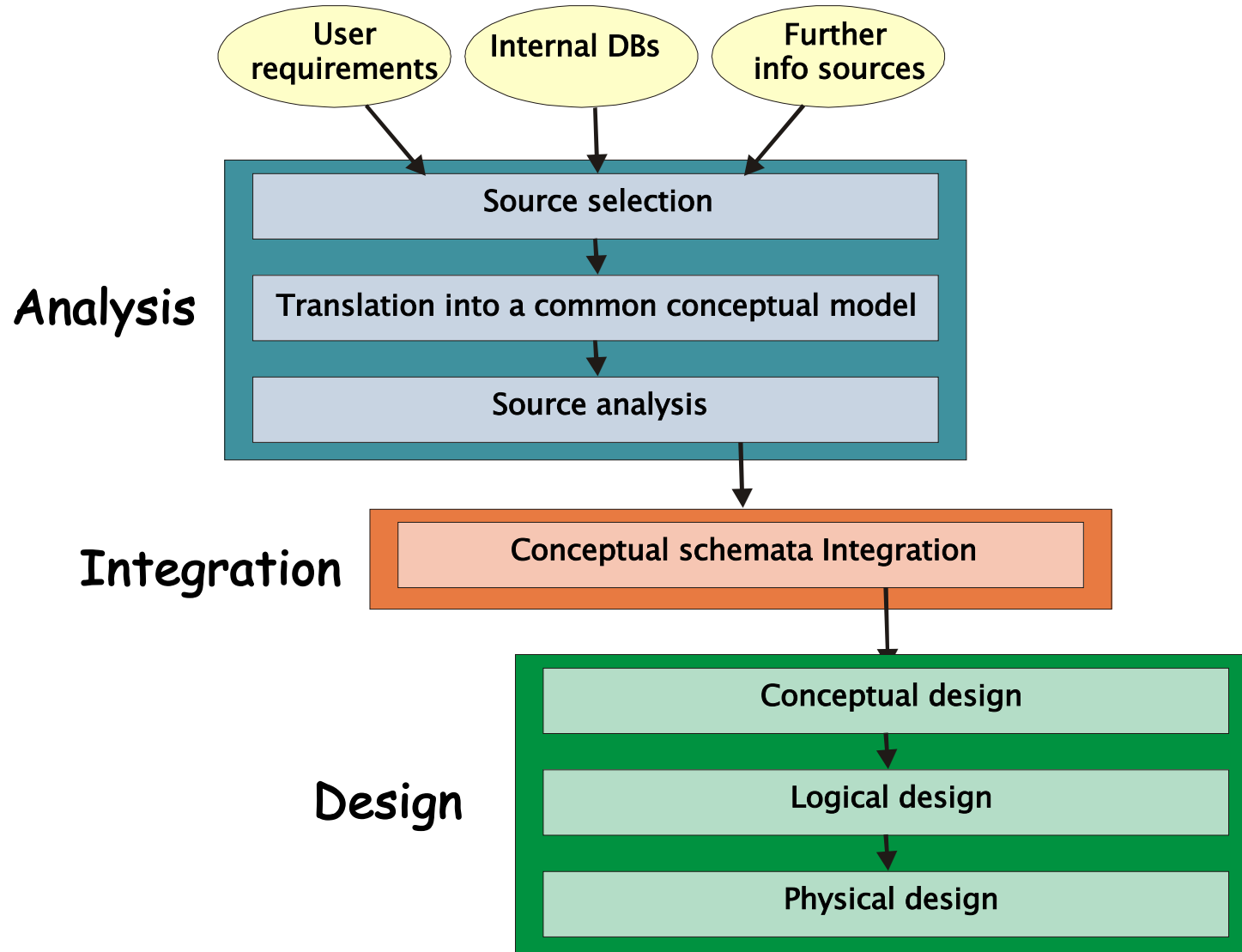


Data Warehouse Design

Letizia Tanca
Politecnico di Milano

with kind thanks to
Rosalba Rossato

Data Warehouse Design

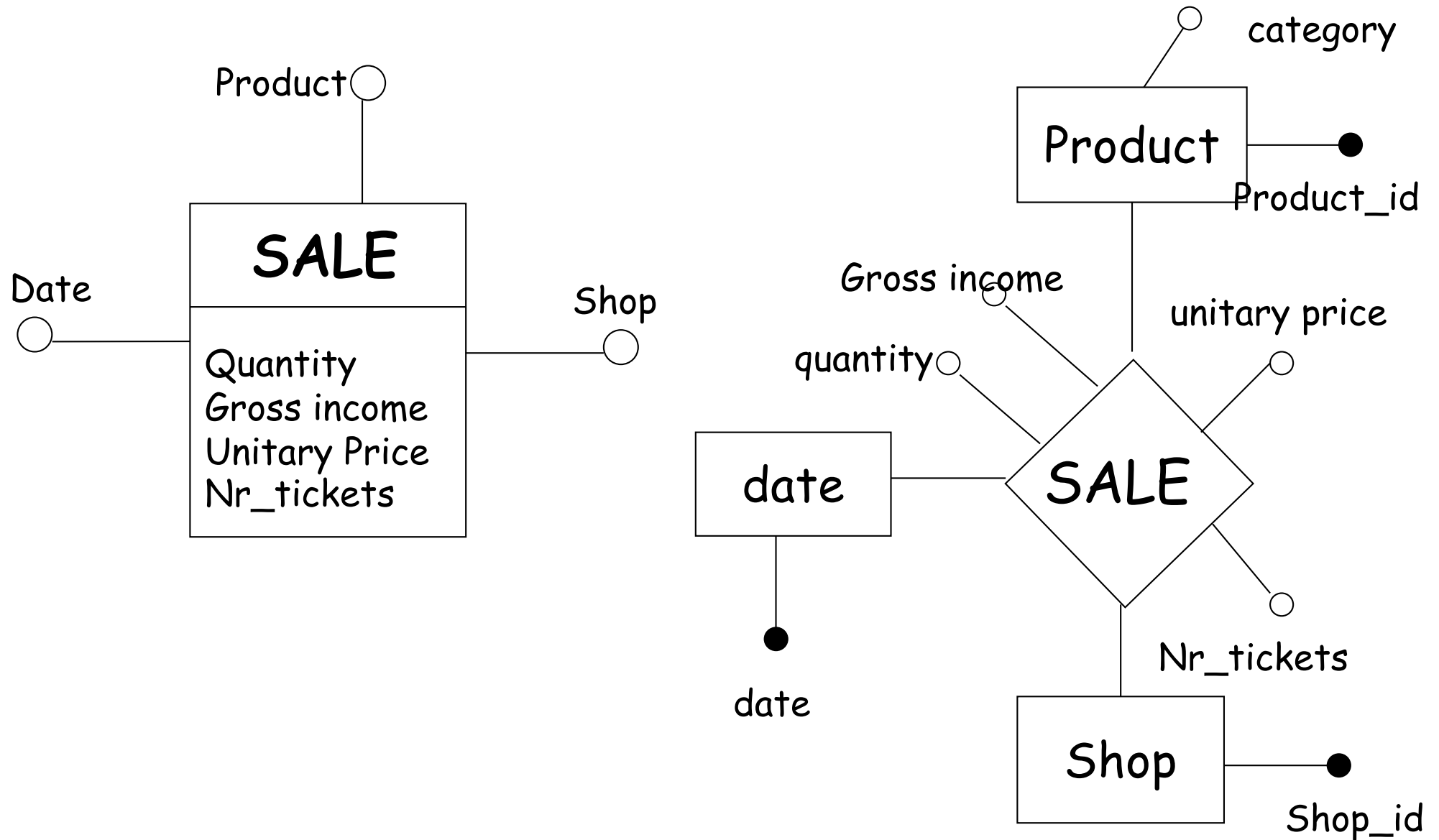


Conceptual design

DW Conceptual design

- DW Conceptual design takes into account the documentation related to the integrated database:
 - Conceptual schema (E/R, UML class diagr.,....)
 - Logical schema (e.g. relational, XML...)

DFM and E/R



Top-down methodology

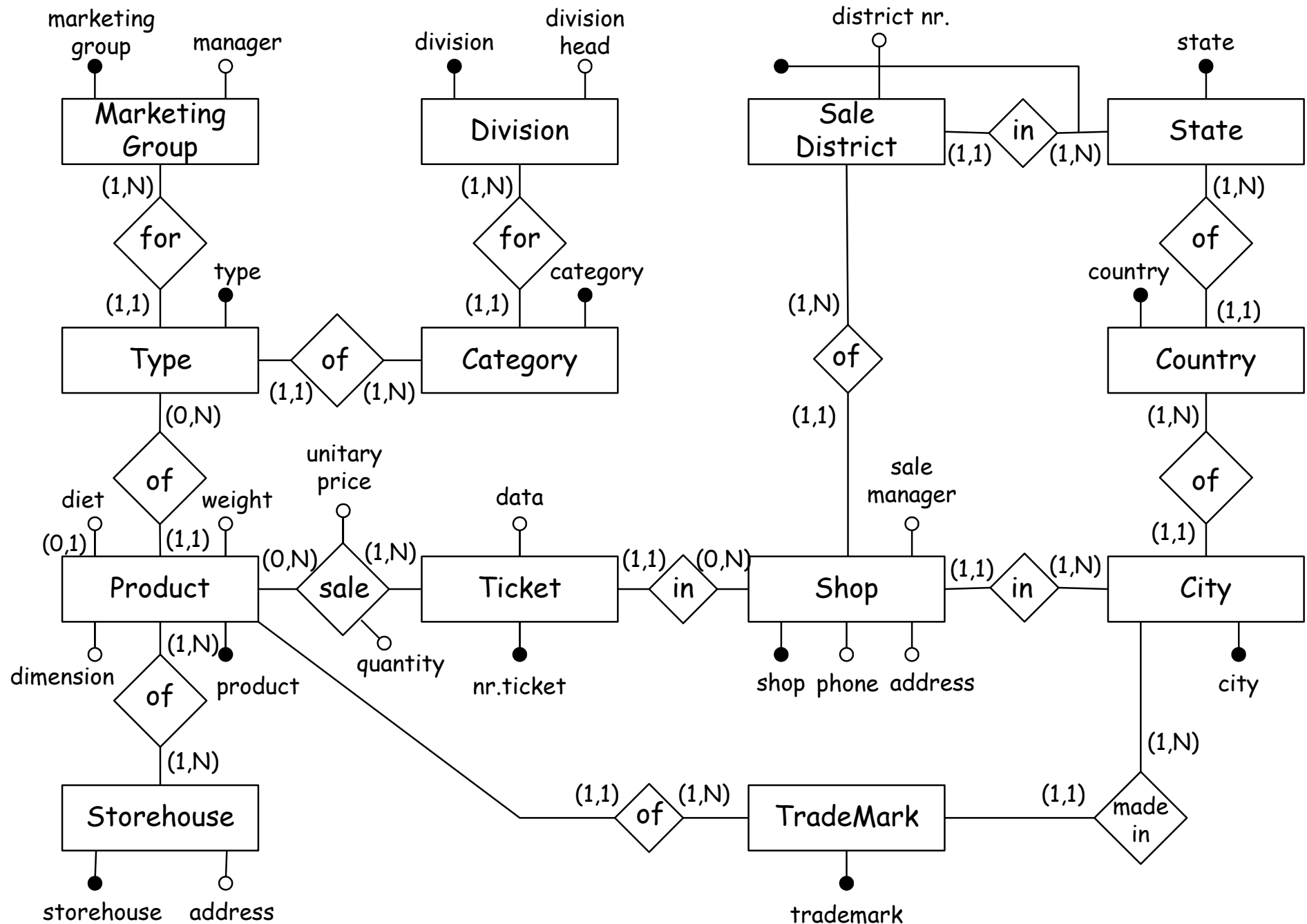
1. Fact choice (**subject oriented !!**)
2. For each fact:
 1. Design of the attribute tree
 2. Attribute-tree editing
 3. Dimension definition
 4. Measure definition
 5. Fact schema creation

Given a Relational Schema

Product(product,weight,dimension,trademark:TradeMark,type:Type)
Shop(shop,address,phone,salemanager,(district,state):District,city:City)
Ticket(nrticket,date,shop:Shop)
Sale(product:Product,nrticket:Ticket,quantity,unitaryprice)
Storehouse(storehouse,address)
City(city,country:Country)
Country(country,state:State)
State(state)
District(district,state:State)
Prod_Storehouse(product:Product,storehouse:Storehouse)
TradeMark(trademark,madein:City)
Type(type,marketinggroup:MarketingGroup,category:Category)
MarketingGroup(marketinggroup,manager)
Category(category,division:Division)
Division(division,divisionhead)

Note: the foreign keys are indicated by quoting the referred table and attribute (s)

We start from the corresponding E/R schema



Fact definition

Facts correspond to **events that dynamically happen in the organization**

- W.r.t. an E/R schema, a fact can correspond to **an entity** F or to **a relationship** among n entities $E_1, E_2, \dots, E_n$
- W.r.t. a relational schema, often a fact corresponds to **a relation (table)** R

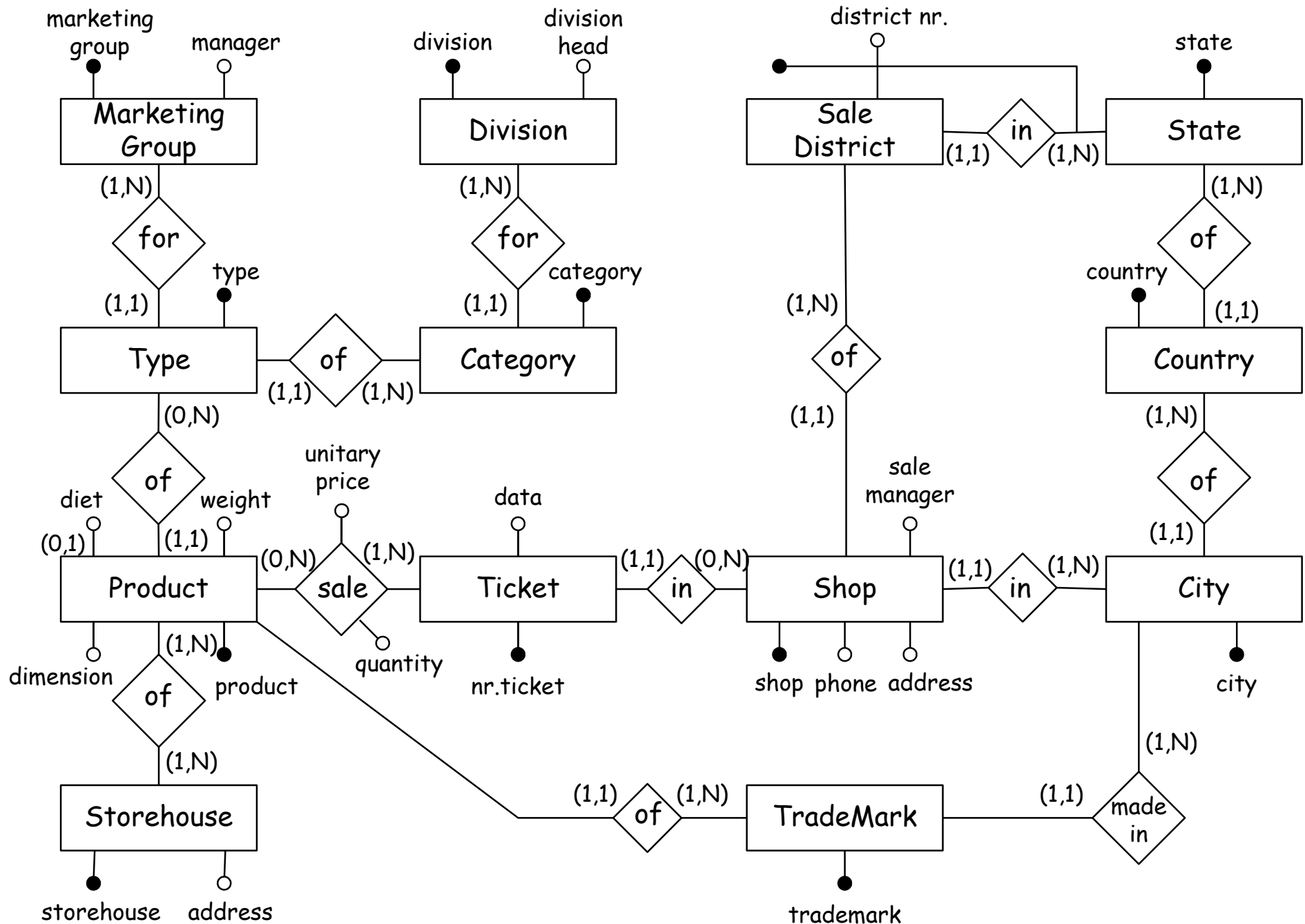
Fact definition

- Good fact candidates: entities or relationships representing frequently updated archives, e.g. "sales"
- Static archives (e.g., "students") are NOT good fact candidates: indeed, in general, a DW is used for studying the dynamics of events (sales, etc...) !!
- Remark: when we identify a fact to be studied, it becomes the root of the fact schema

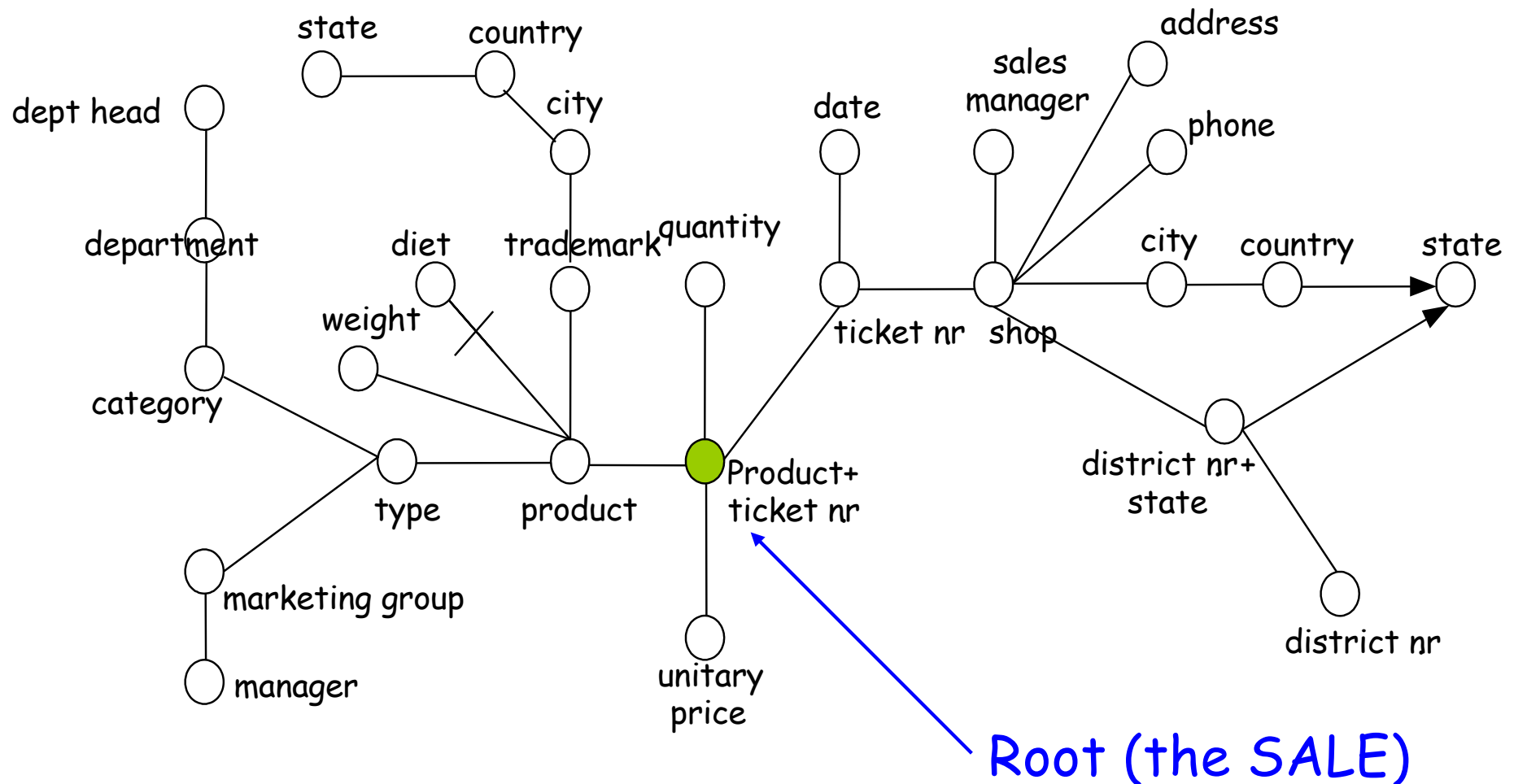
Attribute tree definition

- The **attribute tree** is composed by:
 - **Nodes**, corresponding to attributes (simple or complex) of the source schema
 - **Root**, corresponding to the primary key of the fact we analyze
 - For each node, the corresponding attribute **functionally determines** its descendant attributes
- The attribute tree **can be obtained from the ER schema by using a semi-automatic procedure**

We start from the E/R schema



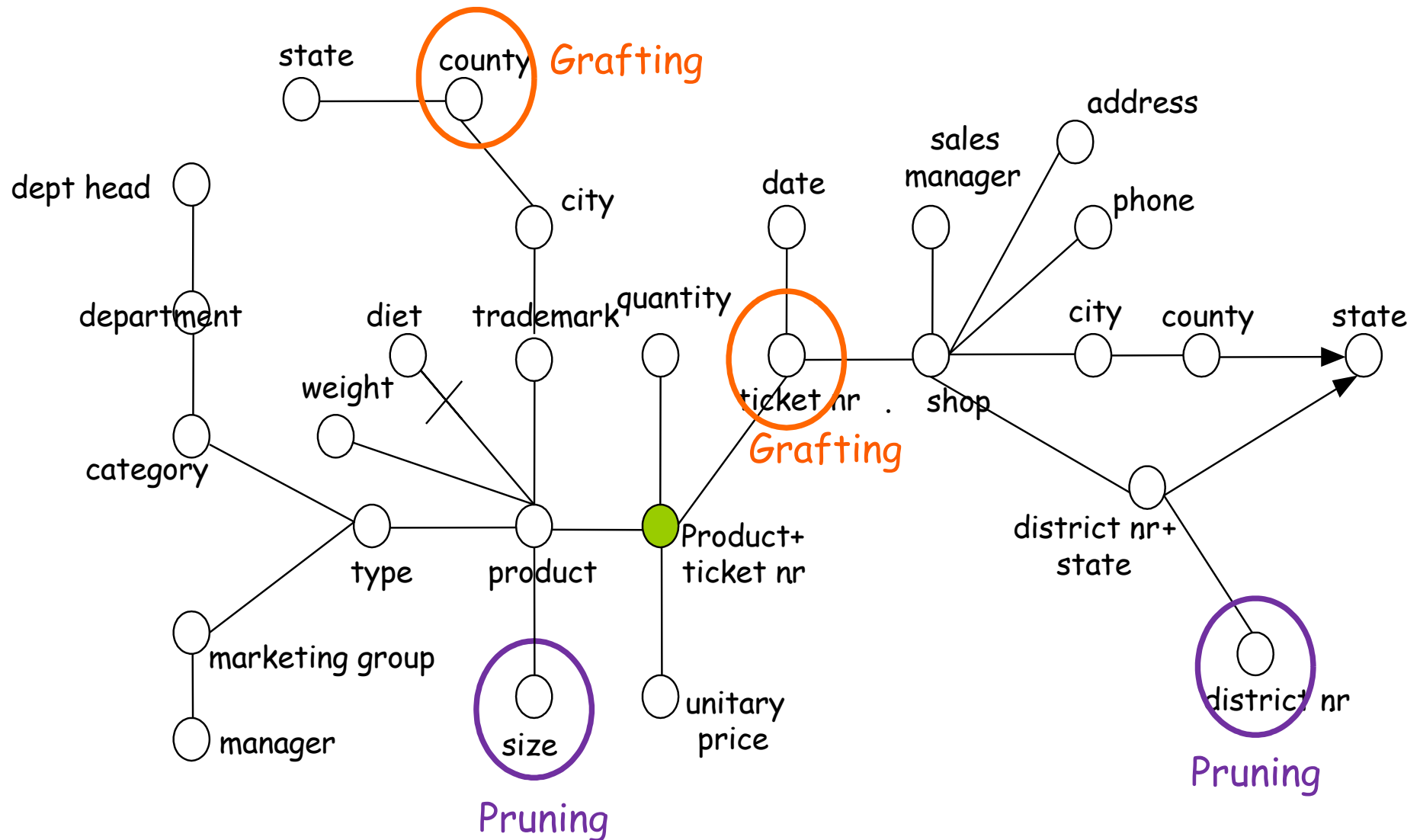
Attribute Tree for the SALE fact



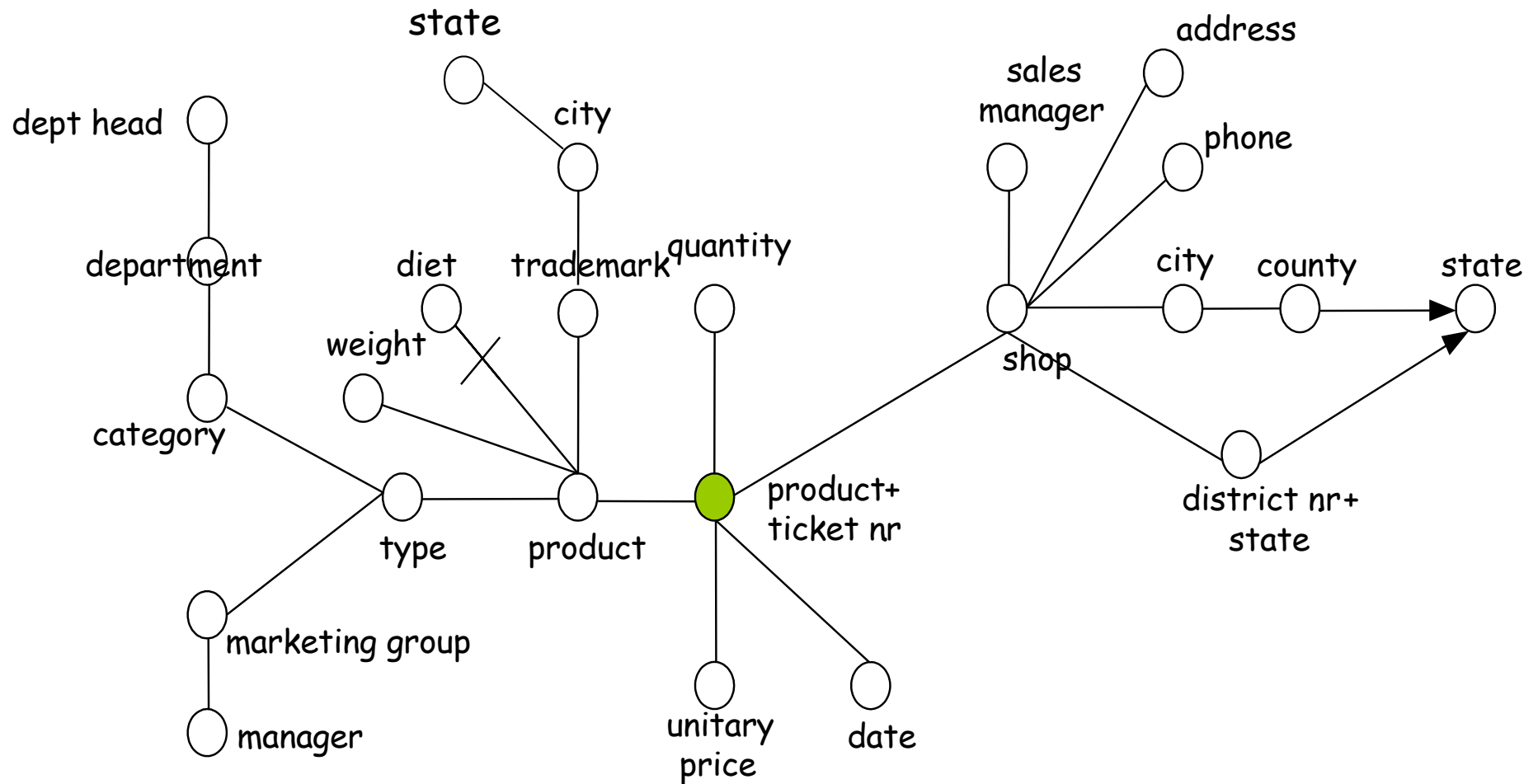
Attribute tree editing

- The editing phase allows to remove those attributes which are irrelevant for the data mart
 - **Pruning of a node v** : the subtree rooted in v is deleted
 - **Grafting of a node v** : the children of v are directly connected to the father of v

Attribute tree editing: example



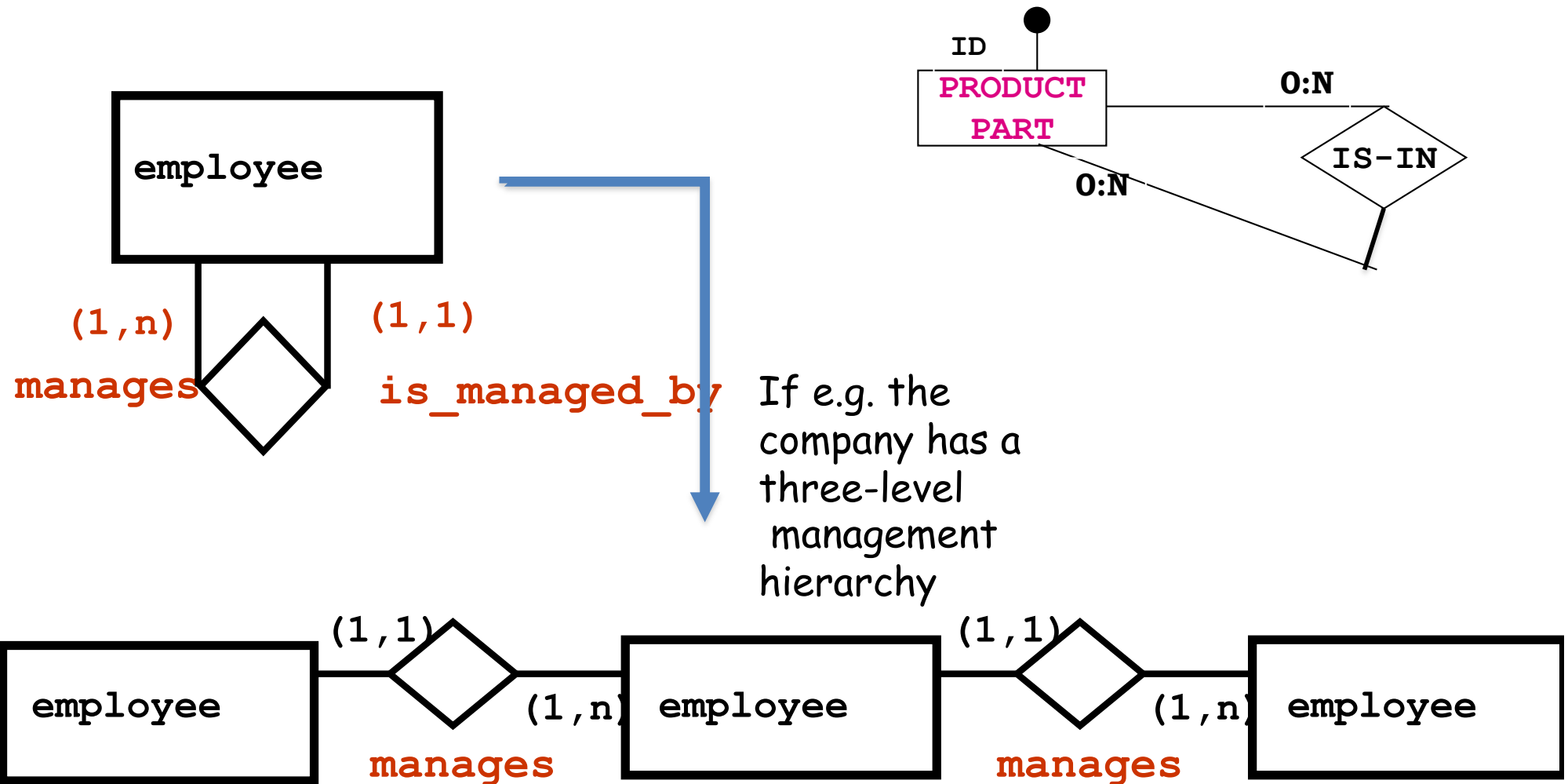
Attribute tree editing: example



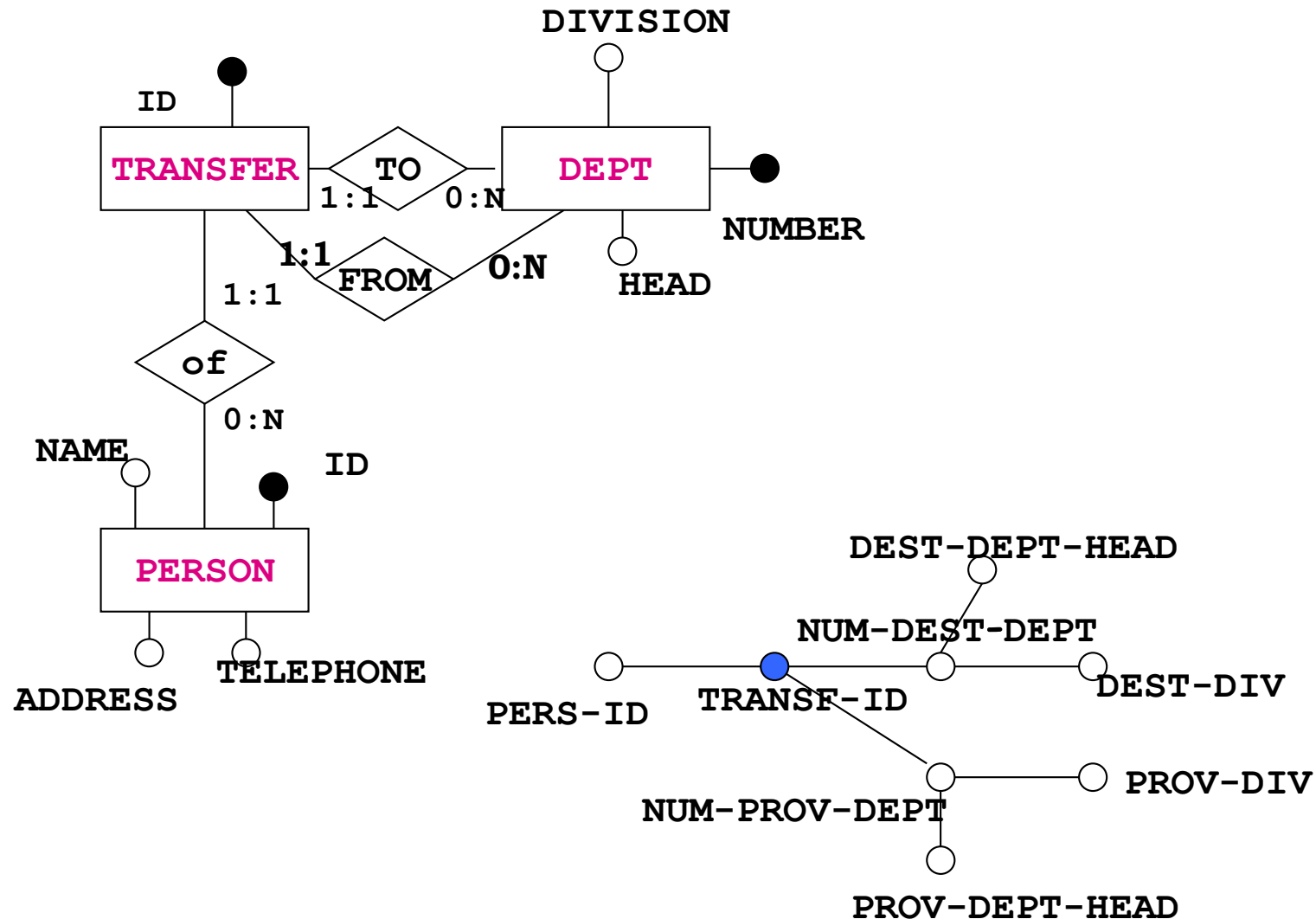
Special cases

- **Cyclic relationships** (e.g. part-subpart, or employee-manager) → they must be broken after a certain number of iterations
- **Cycles in the schema** → they must be broken, possibly choosing to keep the most convenient link
- **ISA hierarchies** → dealt with as optional (0:1) relationships
- **Compound attribute** → vertex with children (the single component attributes)

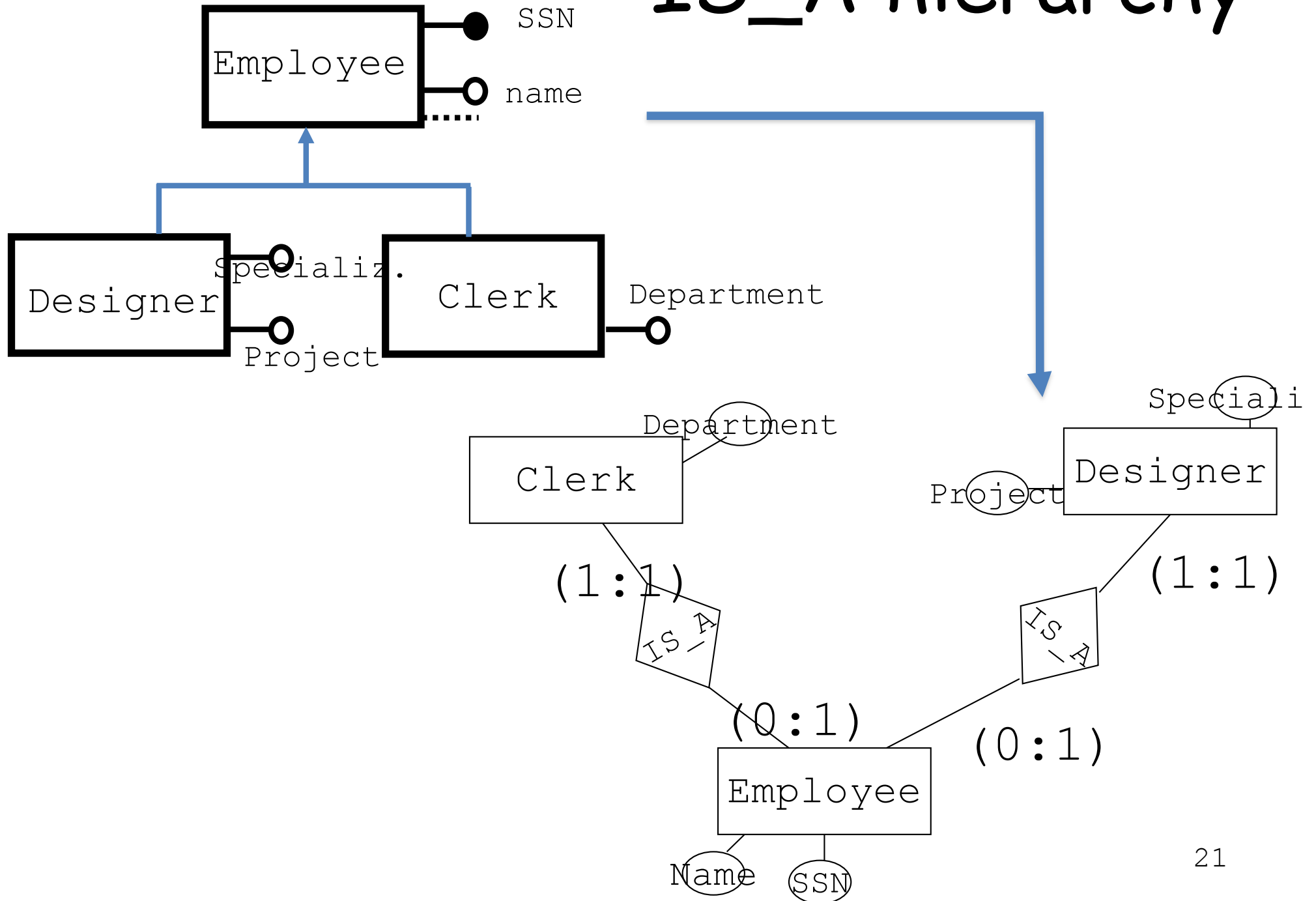
Cyclic Relationships



Cycles in the schema



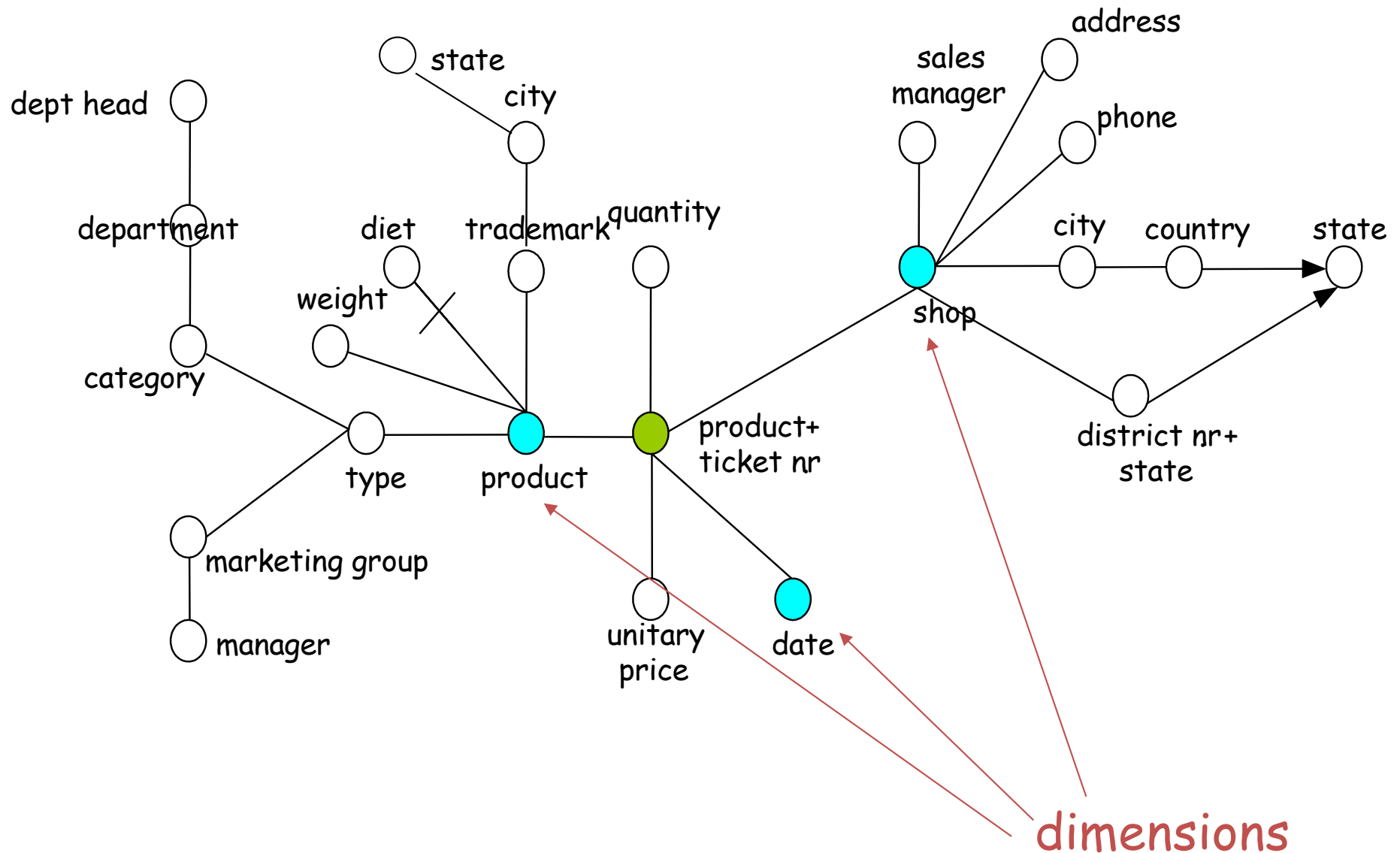
IS_A hierarchy



Dimension choice

- **Dimensions** can be chosen among the attributes that are children of the root of the tree
- **Time should always be a dimension**
 - Historical source: time is an attribute
 - Snapshot source: not always time is directly represented. In this case it is necessary to add time.

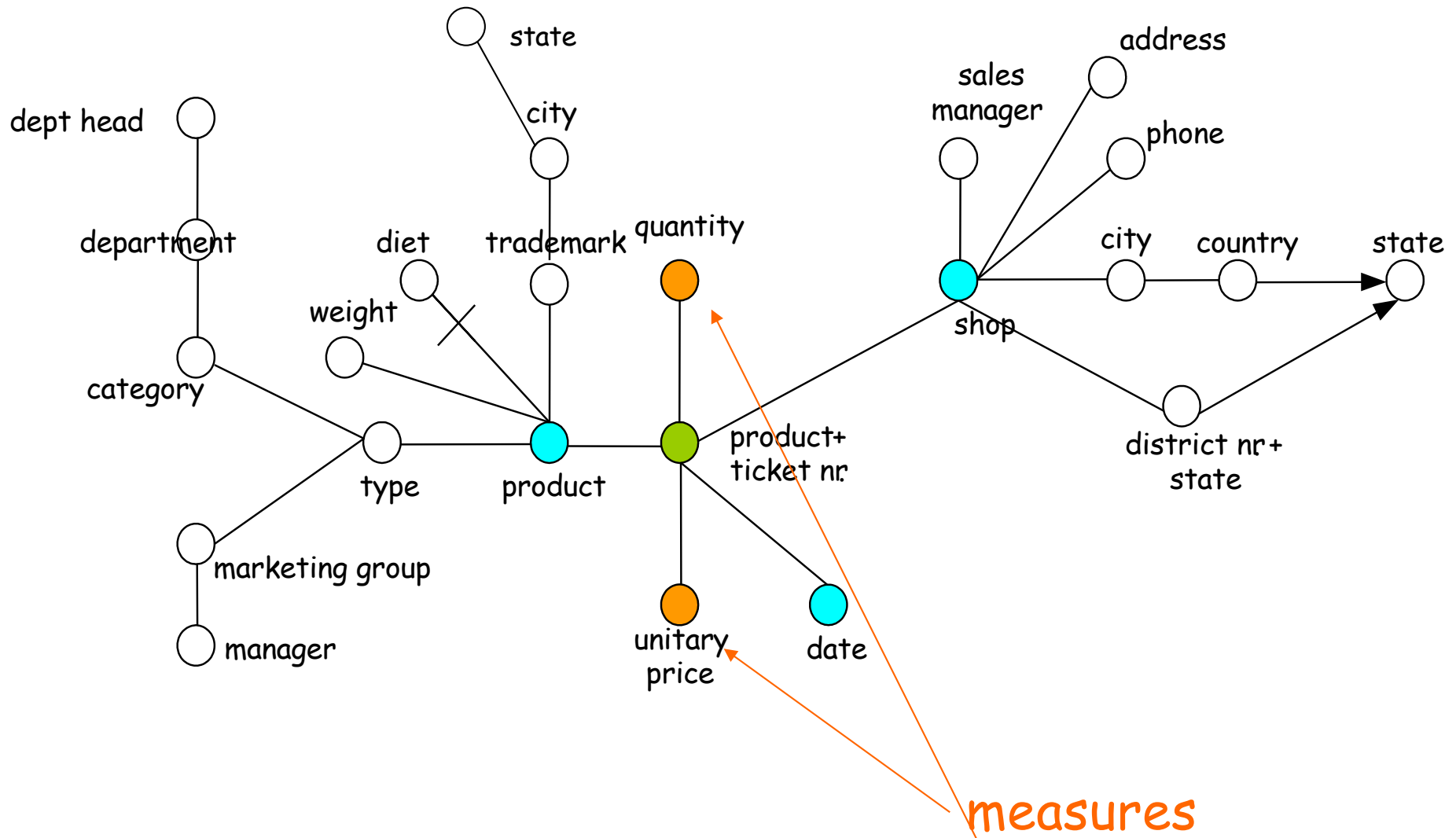
Dimensions definition: example



Measure definition

- While the set of attributes that acts as fact identifier is included in the set of dimensions, numerical attributes of the root (fact) are measures (e.g., unitary price)
- More measures are defined by applying aggregate functions to other attributes of the tree (e.g. ticket count)
- Typical aggregate functions: sum, average, min, max, count

Measure definition: example



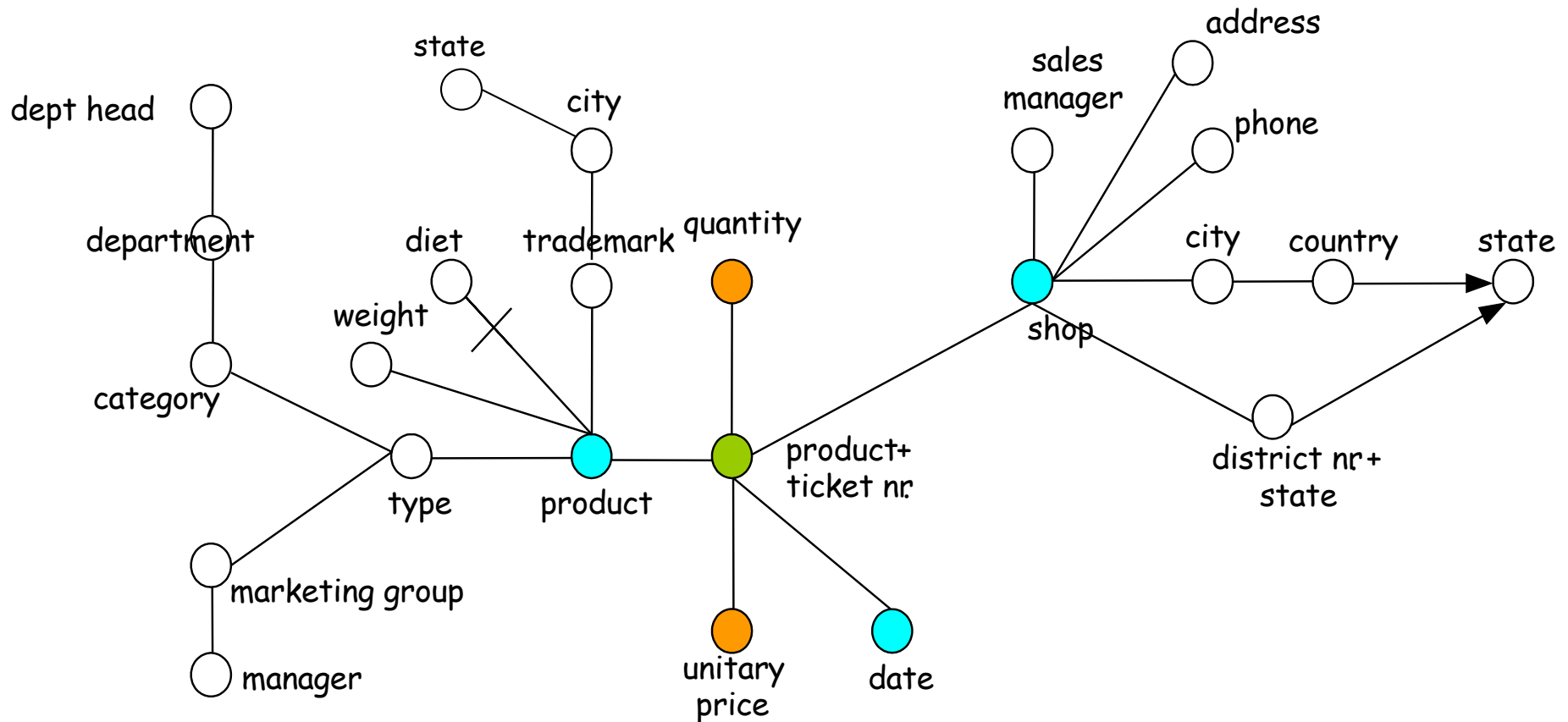
Measure definition (2)

- It is possible that a fact has no measures (empty)
- If the granularity of a fact is different w.r.t. the granularity of the source schema, it can be useful to define suitable measures in order to aggregate the same attribute by using various operators

Glossary

- In the glossary, an expression is associated with each measure
 - The expression describes how we obtain the measure at the different levels of aggregation starting from the attributes of the source schema

Measure definition: example



Quantity = $SUM(\text{Sale.quantity})$

Gross income = $SUM(\text{Sale.quantity} * \text{Sale.unitaryprice})$

Unitary price = $AVG(\text{Sale.unitaryprice})$

Nr-tickets = $COUNT(*)$

Fact schema creation

- The attribute tree is translated into a fact schema including dimensions and measures
 - Dimension hierarchies correspond to subtrees having as roots the different dimensions (with the least granularity)
 - The fact name corresponds to the name of the selected entity

Fact schema creation: example

